Quant Mentorship Final Project

STUDY OVERVIEW

This report investigates how market stress builds and propagates across an equal-weight portfolio of six mega-cap technology stocks: AAPL, MSFT, GOOGL, AMZN, META, and NVDA. Five years of daily OHLCV data (April 2021 to April 2026) were sourced from Yahoo Finance, cleaned to a 1,305-trading-day aligned price matrix, and analyzed using rolling time-series features, unsupervised clustering, and statistical forecasting. The central research question is whether a custom Stress Propagation Score -- built from five daily state metrics -- can meaningfully separate calm, transitional, and stressed market environments, and whether those environments carry materially different risk profiles for portfolio managers.

DATASET

Universe:	AAPL, MSFT, GOOGL, AMZN, META, NVDA
Period:	2021-04-18 to 2026-04-18
Frequency:	Daily adjusted-close OHLCV
Portfolio:	Equal-weight synthetic portfolio
Source:	Yahoo Finance via yfinance (pre-downloaded; no API call required)
Trading days:	1,305 after temporal alignment and forward-fill cleaning
Preprocessing:	Forward-fill at most 2 consecutive gaps (exchange holiday mismatches); then drop rows with any remaining NaN

ANALYTICAL METHODS

Rolling features:	5 state metrics: realized vol, avg pairwise corr, breadth above 50-DMA, dispersion, lead-lag conc.
Stress Score:	Equal-weight z-score blend of the 5 rolling metrics; also included as 6th K-Means clustering feature
Regime detection:	K-Means (k=3, n_init=25, random_state=42) on 6-feature rolling state matrix
Stationarity:	Augmented Dickey-Fuller (ADF) on prices, log returns, realized vol, and stress score
Autocorrelation:	Ljung-Box on portfolio log returns (market efficiency check) and on ARIMA residuals
ARIMA(2,0,1):	20-day realized vol forecast; AIC/BIC model selection; expanding walk-forward evaluation
GBM Monte Carlo:	500 paths, 252-day forward horizon; VaR (95/99%) and CVaR (95/99%) computed

MARKET REGIME SUMMARY

Regime	Days	% of Sample	Avg Vol (20d ann.)	Avg Corr (20d)	Avg Breadth	Avg Stress Score
Calm Expansion	514	40.0%	7.6%	0.14	0.63	-0.93
Rotation / Transition	501	39.0%	17.9%	0.37	0.54	+0.18
Stress Contagion	270	21.0%	45.9%	0.82	0.14	+1.49

REGIME INTERPRETATION

Calm Expansion

Vol: 7.6% | Corr: 0.14 | Breadth: 0.63

Low volatility and high breadth characterize an orderly uptrend. Cross-stock correlation is low (0.14), meaning stocks are largely moving on their own fundamentals rather than in lock-step. Regime persistence is 99.8% per day -- once calm, the portfolio tends to remain calm for extended periods.

Rotation / Transition

Vol: 17.9% | Corr: 0.37 | Breadth: 0.54

Intermediate stress with rising correlation (0.37) and declining breadth (0.54). This regime captures sector rotation, modest pullbacks, and macro uncertainty that has not yet triggered broad contagion. It serves as the leading indicator for Stress Contagion entries, with a 2.8% daily probability of escalating.

Stress Contagion

Vol: 45.9% | Corr: 0.82 | Breadth: 0.14

Sharply elevated volatility (45.90%) with correlation spiking to 0.82 -- stocks move in near lock-step, eliminating the diversification benefit of holding six names. Breadth collapses to 0.14, meaning fewer than 15% of stocks are above their 50-day moving average. Once entered, stress persists 95.2% of trading days.

ARIMA VOLATILITY FORECASTING

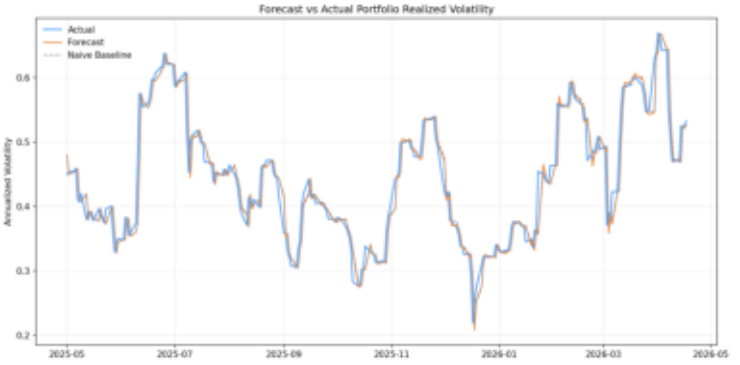
Realized volatility was confirmed stationary by ADF ($p < 0.001$), making it an appropriate ARIMA target. An AIC/BIC grid search over five candidate orders (AR=1-2, I=0-1, MA=0-1) selected ARIMA(2,0,1). Walk-forward evaluation over 250+ expanding-window steps produced RMSE 0.0269 and MAE 0.0167, beating the lag-1 naive baseline (RMSE 0.0271, MAE 0.0167). Ljung-Box tests on residuals confirm white-noise at lags 10 and 20 ($p > 0.05$), validating the model specification. Model fit: AIC = -6592.9, BIC = -6568.2.

KEY FINDINGS

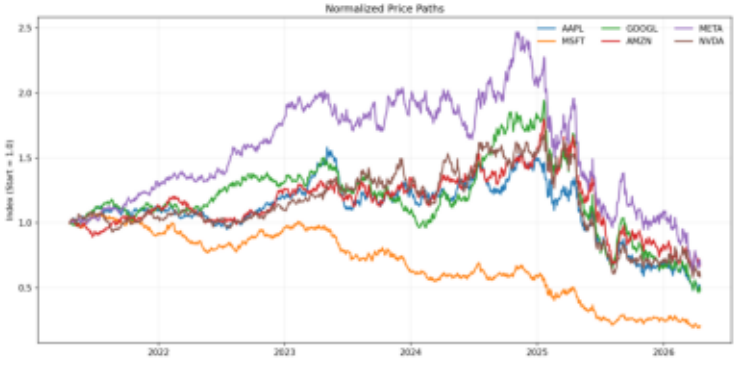
- Stress Contagion carried an average realized vol of 45.90% versus 7.65% in Calm Expansion.
- The Stress Propagation Score was higher in Stress Contagion (+1.49) than in Calm Expansion (-0.93).
- Before stress-regime entries, the score rose in 35.7% of observed transitions.
- The strongest lead-lag relation was GOOGL leading AAPL by 3 day(s) (corr=+0.12).
- ARIMA(2, 0, 1) closely matched the naive volatility baseline (RMSE 0.0269 versus 0.0271).

Stress Propagation Score with Regime Shading

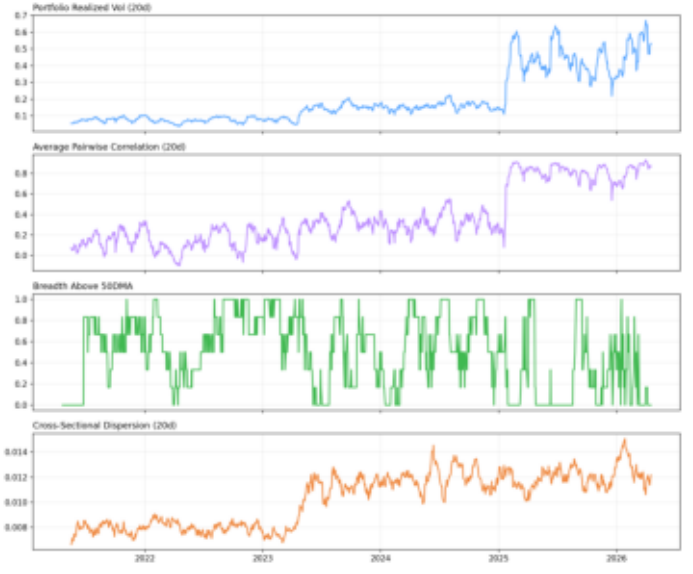
ARIMA(2,0,1): Walk-Forward Forecast vs Actual Vol



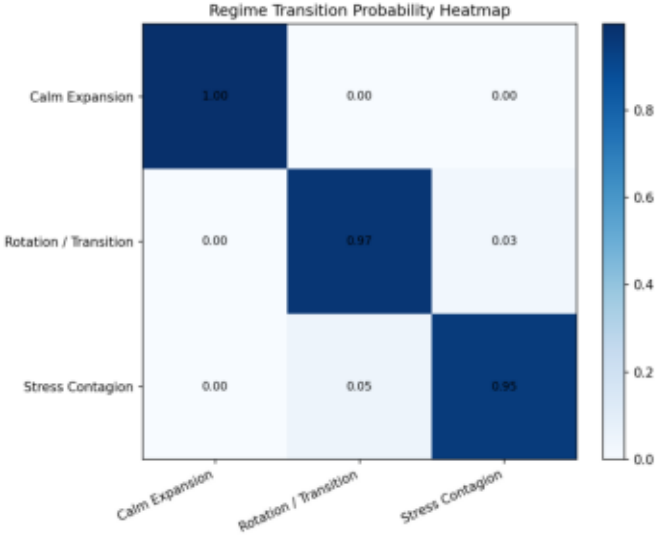
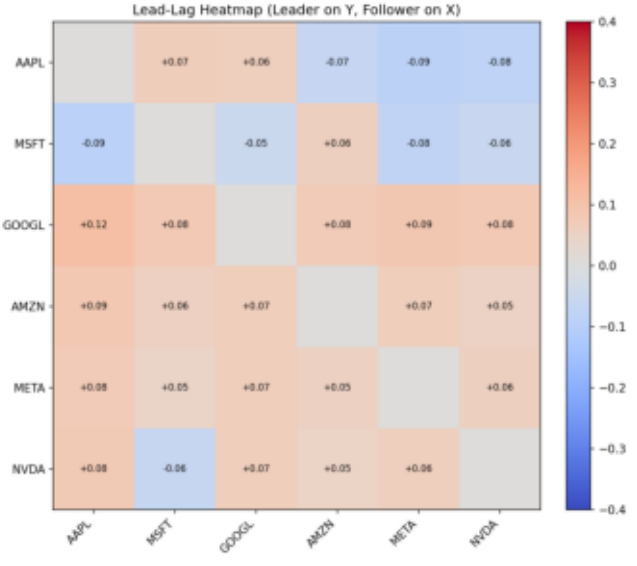
Normalized Price Paths (Base = 1.0 on first day)



Rolling Portfolio State Metrics (20d window)



Regime Transition Probability Heatmap



Lead-Lag Heatmap (Leader on Y, Follower on X)